A DETERMINANT INEQUALITY FOR SUMS OF HERMITIAN EXPONENTIALS

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Abstract. This note records a statement refuted in the *Count ex Machina* archive: each is quoted as posed and answered by an explicit witness, and the archive carries a certificate that recomputes the refutation by rigorous interval arithmetic.

Summary: A proposed determinant inequality comparing sums of Hermitian exponentials. *Certificate:* Integer 3×3 symmetric inputs; exact rational Taylor intervals certify normalized right side $< 1007/1000 < 101/100 <$ normalized left side.

Posed by S. Sra; counterexample found by GPT Astra (Pro) (2026-10-30); formalized by GPT Astra (Pro); audited by GPT Astra (Pro); contributed by S. Sra.

All matrices in a comparison have the same finite order. A Hermitian matrix satisfies $H = H^*$, where $*$ denotes conjugate transpose, and e^H is the usual matrix exponential. The notation $\det(e^H + e^K)^2$ means the square of the determinant, equivalently $\det((e^H + e^K)^2)$, not an entrywise square. For Hermitian H, K, set

$$A = e^H, \qquad B = e^K, \qquad E = e^{H+K}, \qquad D = B^{-1} = e^{-K}.$$

These exponentials are positive definite. Products such as DA need not be Hermitian; the determinant identities used below do not require them to be.

Problem 1 (As posed, paraphrased: S. Sra, Conjecture 4.6, posed in 2015 while studying MathOverflow question 224121 (Wolfgang, 2015))**.** For Hermitian matrices H, K of the same order,

$$\det(e^H + e^K)^2 \le \det(e^{2H} + e^{2K} + 2e^{H+K}).$$

Theorem 2 ([**Disproof**]: Hermitian exponential determinant comparison)**.** *The assertion in Problem 1 is false already for real symmetric* 3×3 *matrices.*

Proof. Take the exact integer matrices

$$H = \begin{pmatrix} -26 & 10 & -10 \\ 10 & -16 & 20 \\ -10 & 20 & -26 \end{pmatrix}, \qquad K = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 20 \end{pmatrix}.$$

Both are real symmetric, hence Hermitian, of the same order. Define

$$L = \frac{\det(A+B)^2}{\det(B)^2}, \qquad R = \frac{\det(A^2 + B^2 + 2E)}{\det(B)^2}.$$

Here $\det(B)^2 = e^{2\,\mathrm{tr}\,K} = e^{60} > 0$, so the proposed inequality is equivalent to $L \le R$. Left multiplication and congruence, respectively, give the exact identities

$$L = \det(I + DA)^2, \qquad R = \det(I + DA^2D + 2DED).$$

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From the *Count ex Machina* archive of counterexamples, https://github.com/suvrit/count-ex-machina, whose companion paper is arXiv:2608.29595.

The supplied verifier certifies

$$1.010625961431267059748845465 7 < L < 1.010625961431267059748845465 8,$$
$$1.006636966146830492353750702 0 < R < 1.006636966146830492353750702 1.$$

Every terminating decimal in these strict bounds denotes an exact rational number, not a rounded estimate. In particular,

$$R < \frac{1007}{1000} < \frac{101}{100} < L, \qquad L - R > \frac{3}{1000}.$$

This reverses the claimed comparison.

For completeness, here is the error control behind the certificate. For a rational matrix X put

$$r = \|X\|_\infty = \max_i \sum_j |X_{ij}|, \qquad T_m(X) = \sum_{j=0}^{m} \frac{X^j}{j!}.$$

Submultiplicativity and the power-series definition imply

$$\|e^X - T_m(X)\|_\infty \le \sum_{j=m+1}^{\infty} \frac{r^j}{j!} \le \frac{r^{m+1}}{(m+1)!} \frac{1}{1 - r/(m+2)} =: \varepsilon_m(r), \qquad r < m + 2.$$

The last bound is a geometric-series bound: after the first omitted term, each successive ratio is at most $r/(m+2)$. Therefore every entry of e^X lies in the rational interval $[(T_m(X))_{ij} - \varepsilon_m(r), (T_m(X))_{ij} + \varepsilon_m(r)]$.

The verifier takes $m = 256$. The required norms are $\|H\|_\infty = 56$, $\|H + K\|_\infty = 46$, and $0, 10, 20$ for the three scalar exponents forming D. Each is strictly below 258. The off-diagonal entries of D are exactly zero. Rational interval addition, subtraction, and multiplication propagate these enclosures through the two normalized determinant expressions, using the six-term determinant formula in order three. Before squaring the left determinant interval, its strict positivity is certified. Interval endpoints and every inequality test use integers and `fractions.Fraction`; there is no unbounded truncation error and no machine-precision comparison. All displayed enclosing bounds and the strict separation above are checked, not merely printed. □

Remark 3 (The linear coefficient and a log-determinant normalization). The following coefficientwise strengthening is recorded only as an auxiliary observation, not as a second archival result. For fixed A, B, E, let

$$\Delta(t) = \det(B^2 + 2tE + t^2A^2) - \det(B + tA)^2.$$

Jacobi's determinant differentiation formula at $t = 0$ gives

$$\frac{[t]\Delta(t)}{\det(B)^2} = 2\big(\operatorname{tr}(B^{-2}E) - \operatorname{tr}(B^{-1}A)\big) = 2\big(\operatorname{tr}(DED) - \operatorname{tr}(DA)\big).$$

For the same exact witness, the verifier certifies that this number c_1 satisfies

$$-0.008112047833566766616325867 6 < c_1 < -0.008112047833566766616325867 5 < -\frac{1}{125} < 0.$$

Thus even the linear coefficient has the wrong sign.

This is also the obstruction exposed by a log-determinant normalization. Put $S_t = B^2 + t^2A^2$. Since $(B + tA)^2 = S_t + t(AB + BA)$, cancellation of $\det S_t$ gives, for t near zero,

$$\begin{aligned} F(t) &= \operatorname{tr}\log\big(I + 2tS_t^{-1/2}ES_t^{-1/2}\big) \\ &\quad - \operatorname{tr}\log\big(I + tS_t^{-1/2}(AB + BA)S_t^{-1/2}\big) \\ &= c_1 t + O(t^2). \end{aligned}$$

The logarithms are defined in a neighborhood of zero by positivity and continuity. The expansion is used to explain the discovery, not as a substitute for the exact certificate at $t = 1$.

Remark 4 (An analytic family explaining the failure). A separate limiting construction explains why counterexamples exist. Set

$$M = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 1 & 2 \\ -1 & 2 & 0 \end{pmatrix}, \qquad N = \operatorname{diag}(0,1,2), \qquad \lambda = \lambda_{\max}(M).$$

The unit vector $v = (0,1,1)^T/\sqrt{2}$ has $v^T M v = 5/2$, so $\lambda \geq 5/2 > 2$. No eigenvector of M can have first coordinate zero: writing its eigenvalue as μ, the first row would give $u_2 = u_3$, and the second and third rows would then give simultaneously $\mu u_2 = 3u_2$ and $\mu u_2 = 2u_2$. This would force the entire vector to vanish. Consequently each eigenspace is one-dimensional (otherwise a linear combination would have first coordinate zero). Let $P = uu^T$ be the rank-one top spectral projection, with $\|u\| = 1$, and put $Q = e_1 e_1^T$, $\alpha = |u_1|^2 > 0$.

For $r > 0$ define

$$H_r = r(M - \lambda I), \quad K_r = rN, \quad A_r = e^{H_r}, \quad B_r = e^{K_r}, \quad D_r = B_r^{-1}, \quad E_r = e^{H_r + K_r}.$$

Then $A_r \longrightarrow P$ and $D_r \longrightarrow Q$. The matrix $T = M + N$ has characteristic polynomial $(x-4)(x^2-2)$ and top unit eigenvector v. Since this vector has zero first coordinate,

$$\begin{aligned} \|D_r E_r D_r\|_2 &\leq e^{(\sqrt{2}-\lambda)r} + e^{(4-\lambda)r}\|D_r v\|^2 \\ &= e^{(\sqrt{2}-\lambda)r} + \tfrac{1}{2}e^{(2-\lambda)r} + \tfrac{1}{2}e^{-\lambda r} \longrightarrow 0. \end{aligned}$$

Here $\|\cdot\|_2$ is the operator norm. The contribution from the two remaining eigenspaces is bounded by $e^{(\sqrt{2}-\lambda)r}$, since $\|D_r\|_2 = 1$.

It follows that, coefficientwise as polynomials in t,

$$\begin{aligned} \frac{\det(B_r^2 + 2tE_r + t^2 A_r^2)}{\det(B_r)^2} &= \det(I + 2tD_r E_r D_r + t^2 D_r A_r^2 D_r) \longrightarrow 1 + \alpha t^2, \\ \frac{\det(B_r + tA_r)^2}{\det(B_r)^2} &= \det(I + tD_r A_r)^2 \longrightarrow (1+\alpha t)^2. \end{aligned}$$

Coefficientwise convergence follows because these coefficients are polynomial functions of the converging matrix entries. The normalized difference tends to

$$-2\alpha t + \alpha(1-\alpha)t^2.$$

At $t = 1$ its limit is $-\alpha(1+\alpha) < 0$, so the original inequality fails for all sufficiently large r. Its linear coefficient tends to $-2\alpha < 0$. This analytic explanation is not needed to validate the fixed integer witness above; in particular, no numerical value of λ enters the executable certificate.

Remark 5 (Scope). The certified conclusion concerns the displayed universal assertion for same-order Hermitian matrices, with the direction $\leq$ and with the term $2e^{H+K}$ unchanged.

Commuting pairs satisfy equality, and this case does not dispute that fact. No claim is made that dimension three is minimal, nor is the dimension-two case resolved here. The negative coefficient above is auxiliary, not a separately provenanced claim. No novelty or priority is asserted.

Code and certificate

The case is the directory `counterexamples/hermitian-exponential-determinant/` of the archive, at commit `782fc21`. Its `verify.py` recomputes every refutation in this note using exact arithmetic or rigorous interval enclosures, never floating-point evidence alone; it raises an error if any check fails and otherwise writes the certificate below. To rerun it:

```
git clone https://github.com/suvrit/count-ex-machina
cd count-ex-machina/counterexamples/hermitian-exponential-determinant
python verify.py
```

- Theorem 2 (result `hermitian-exponential-determinant`): rigorous interval arithmetic.
- `artifacts/certificate.json`: Integer witness, rational Taylor-error bounds, outward-rounded rational enclosures, and strict separating inequalities.

AI USAGE

The credit line under the title records, for each statement, who posed it, who found its counterexample and when, and who formalized and audited it; the counterexample here was found by GPT Astra (Pro) (2026-10-30). This note was assembled mechanically from the case's source in the archive by `tools/notes.py`, written with Claude Code; its mathematics is exactly that of the archive at the commit cited above.

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REFERENCES

Wolfgang. Which ordering of factors is needed to obtain this kind of determinantal inequalities? MathOverflow question 224121, November 2015. URL https://mathoverflow.net/q/224121. Posed 20 November 2015; unanswered as of 30 October 2026.